

CY 645: Reagents in organic synthesis

Course Code	CY 645
Course Name	Reagents in organic synthesis
L-T-P-C	3-0-0-3
Prerequisites	Basic knowledge in organic reaction and mechanism
Intended for	PhD
Distribution	Elective
Approval	4th Senate (May 2013)
Discontinued	60th BoA (Jun 2025)

Course Contents

- **Introduction**
Classification of reagents; about air and moisture sensitive reagents; handling, storage and precaution; (3 h) Oxidising and reducing agents Synthesis of various oxidizing and reducing reagents; application in the synthesis of natural products, in organic transformation and in asymmetric synthesis [7 Lectures]
- **Solid supported reagents**
Advantages and disadvantages; various inorganic supports and their application in organic reaction; effect of support materials on reactivity; organic supports including polymeric resins, their functionalization and application. [7 Lectures]
- **Hypervalent iodine reagents**
Various types of hypervalent iodine reagents and their preparation; application in organic transformation, selectivity, sensitivity and reactivity [4 Lectures] Peptide coupling reagents Phosphonium, immonium, imidazolium, carbodiimide, uronium, organophosphorous, chloroformate and acid halogenating reagents; their applications. [5 Lectures]
- **Functional group protecting agents**
Different types of protecting/masking agents; protection of chemical functionalities and their deprotection after completion of reaction; their application in chemo- and regioselective reaction (5 h) Copper, palladium, platinum and ruthenium based coupling reagents Preparation of Copper, palladium, platinum and ruthenium based coupling reagents; application in C-C, C-N, C-O and C-S bond forming reactions; their applications in asymmetric synthesis. [5 Lectures]
- **Lanthanide reagents in organic synthesis**
Applications in oxidation, reduction, C-C bond forming, cyclopropanation and cycloaddition reactions. [2 Lectures]
- **P, S, Se and Si-based reagents**
Synthesis and application in organic synthesis. [3 Lectures]

Text Books

1. Bradford P. Mundy, Michael G., **Reactions and Reagents in Organic Synthesis**.
2. Richard F HeckFavaloro, **Palladium reagents in organic syntheses**.

References

1. **Encyclopedia of Reagents for Organic Synthesis**
2. **Fieser and Fieser's Reagents for Organic Synthesis**
3. Anthony J. Pearson, William R. Roush, **Handbook of Reagents for Organic Synthesis, Activating Agents and Protecting Groups**.
4. David Crich, **Handbook of Reagents for Organic Synthesis: Reagents for glycoside, nucleotide, and peptide synthesis**.
5. Philip L. Fuchs, **Handbook of Reagents for Organic Synthesis: Reagents for direct functionalization of C-H bonds**.
6. Leo A. Paquette, **Fluorine-Containing Reagents**
7. Peter Wipf, **Handbook of Reagents for Organic Synthesis: Reagents for high-throughput solid-phase and solution-phase organic synthesis**.
8. Theodora W. Greene, Peter G. M. Wuts, **Protective groups in organic synthesis**
9. **Review articles on reagents from leading journals of organic chemistry.**